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Abstract

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1 Introduction

[This is only a skeleton of the intro] Much of the world knowledge can be expressed in terms of relations, but oftentimes such relations are graded, implied, and nonliteral. Such kind of implicit inference of event relations is common in human language processing... (cite) Talk about eliciture and why its implied relationships between any two events are non-mandated. (cite)

Previous studies have attempted to show how LMs can build a world model through pragmatic inference... (cite) To the best of our knowledge, Pan & Kehler is the only work that examines LM's ability to perform eliciture inference. They demonstrate that LMs not only draw eliciture but also leverage eliciture to guide downstream sentential processing. we aim to extend on their findings by evaluating models of different families and studying the internal representations of implicit causal inference.

The benefits of understanding elicitors in LMs are multifold: 1. whether LMs–trained with distributional objective–demonstrate human-like nonliteral inference in non-mandated setting 2. more comprehensive views into how LMs build an internal world model 3. Potentially understand the implicit bias in LMs, which has implications in safety and alignment

To these aims, we explore three main questions:

RQ1 *Do model behaviors reflect inferences about implicit causal relationships within sentences?*
RQ2 *Do LMs form internal representations of implicit causal relationships?*
RQ3 *If **RQ2** is true, do representations of implicit causality generalize to naturalistic settings with potential real-world implications?*

Our contribution is...

2 Background

3 Experiments

3.1 Measuring Implicit Causal Inference through Model Behaviors

Models. In all experiments, we test Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024] base checkpoints.

Dataset. The dataset tests whether models represent implicit causal relationships. It is comprised of minimal pairs of sentences following the format of [person] [modifier/relative clause] [outcome]. In a critical sentence, [modifier/relative clause] provides a plausible cause of [outcome] based on physical or everyday world knowledge; in a *control* sentence, [modifier/relative clause] provides an incidental detail. For example, *The executive who falsified quarterly revenue figures was fired* is paired with

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The executive who wore a blue silk tie was fired. We use 150 manually authored pairs of sentences. See Appendix A for qualitative examples.

Yes/No. We first present the sentences of [person] [modifier/relative clause] [outcome] as context and then ask, "Does [modifier/relative clause] explain why [outcome]?" We compute the probabilities of the complete answers *Yes* and *No*, including capitalization variants, and normalize them to obtain $P(\text{Yes})$. See Appendix A for an example prompt.

7-point Likert. We present sentences of [person] [modifier/relative clause] [outcome] as context and then ask how strongly the [modifier/relative clause] explains the [outcome] on a scale from 1 (definitely not) through 4 (uncertain) to 7 (definitely yes). We compute a probability for each of the seven complete answers and use their probability-weighted mean as the model's rating. See Appendix A for details.

Surprisal. Following Pan and Kehler [2025], we remove the sentence-final period and append *", and I don't know why."* to the end of a tested sentence. We use the surprisal of this continuation as the score. The intuition is that if [modifier/relative clause] has already supplied an explanation, subsequently claiming *I don't know why* should be less expected and therefore more surprising. More details can be found in Appendix A.

For all three measures, paired accuracy is the proportion of pairs in which the critical sentence scores strictly above its control (by rating or surprisal); chance is .5, ties count as incorrect.

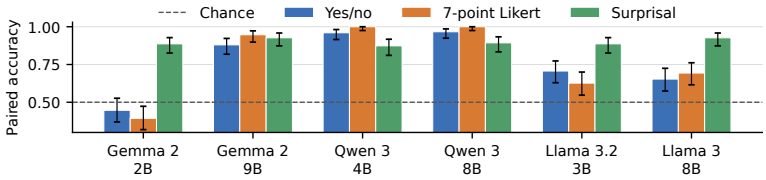


Figure 1: Paired accuracy of model judgments on implicit causal inference items, comparing Yes/No prompting, 7-point likert rating, and the surprisal of the continuation *", and I don't know why."* Higher accuracy indicates better performance in implicit causality inferences.

Results. We present the behavioral evaluation of implicit causal inference in Figure 1. As shown in the figure, models demonstrate implicit causal inference most consistently through surprisals. Not all models can faithfully report their implicit causal inference through Yes/No and 7-point likert rating, and the performance of such metalinguistic causal inference increases with model size.

3.2 Identifying Internal Representations of Implicit Causality Inference

Given that LMs behaviorally demonstrate implicit causal inference, do LMs form internal representations of implicit causality? Specifically, we ask two questions:

1) Do model activations distinguish implicit causal relationships from other relations? If a representation is specific to causality, a causal direction should separate causal pairs well but transfer less reliably to occasional or plausibility pairs.

2) If such representations exist, could the representations of cross-sentence implicit causality transfer to within-sentence causality? Within-sentence causal inferences through relative clauses or modifiers are less strong and mandated compared to cross-sentence causal inferences. If the representation of implicit causality inference is robust, the representation should transfer across syntactic variations.

Datasets. To measure the transferability of implicit causality representations, we use three datasets of matched critical/control pairs:

1) *Causal*: This dataset tests whether models represent implicit causal relationships across sentences. We construct 960 pairs in which an outcome is followed by either a potential explanation (critical) or an incidental detail (control), separated by a period or semicolon. For the cross-transfer analysis, we use the 480 period-separated pairs, with 384 for CAA training and 96 for in-distribution evaluation. We evaluate syntactic generalization on the 150 within-sentence pairs introduced in subsection 3.1. See Appendix B for an example.

2) *Occasion*: This dataset tests whether an event is compatible with a person's typical activity. We adapt and expand the occasion items from Pan and Kehler [2025]. It contains 1,200 critical/control pairs: we use 800 for CAA training and 200 for in-distribution evaluation, and reserve 200 for future testing. See Appendix B for an example.

3) *Plausibility*: This dataset tests whether models distinguish probable events from physically impossible events. We adapt the probable–impossible pairs from Hu et al. [2025], following Lepori et al. [2026]. It contains 70 pairs, with 56 for CAA training and 14 for in-distribution evaluation. See Appendix B for an example.

CAA directions. Contrastive Activation Addition (CAA) represents a relation with a direction in a model's activation space [Rimsky et al., 2024]. At each tested layer, we subtract the final-token activation of every control item from that of its matched critical item, then average these differences over the training pairs. An evaluation item receives a scalar score by projection onto this direction of averaged difference; paired accuracy is the proportion of evaluation pairs for which the critical item scores higher.

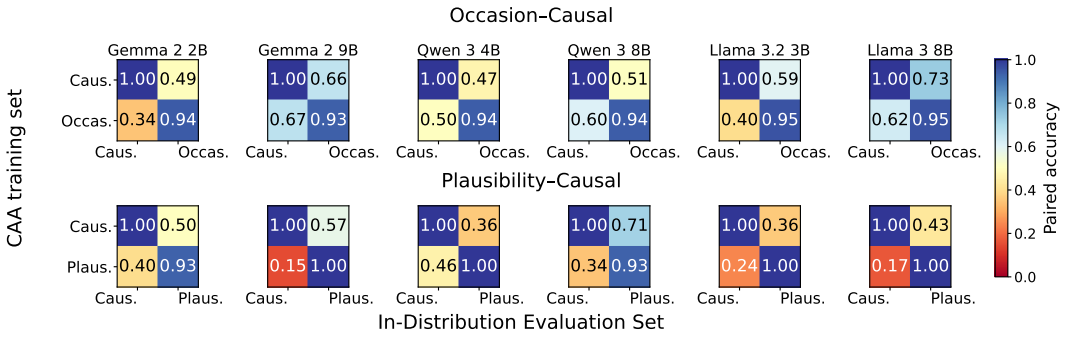


Figure 2: Cross-transfer paired accuracy between causal and occasional relations (top row) and causal and plausibility relations (bottom row). Within each heatmap, rows show the CAA training set and columns show the in-distribution evaluation set. **Causal CAA directions are largely relation-selective and distinct from occasional and plausibility directions.**

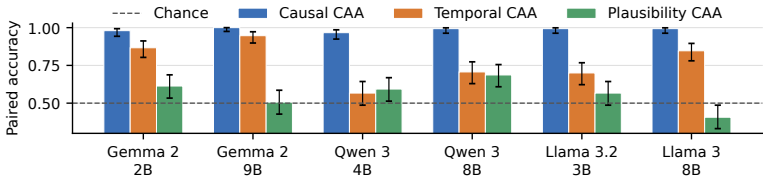


Figure 3: Paired accuracy of causal, occasional, and plausibility CAA directions on 150 unseen within-sentence causal inference pairs with 95% CIs. **Causal CAA directions generalize consistently to within-sentence implicit causal inference.** occasional and plausibility CAA directions transfer less consistently to within-sentence implicit causal items.

Results.

3.3 Generalization to Naturalistic Scenarios

Dataset.

Table 1: Mean paired differences in causal CAA projection in naturalistic causal scenarios. Positive values indicate that a demographic group has a greater mean projection than the white-male condition. Bolded values indicate statsitically significant difference from the white-male projection scores. **Demographic descriptions are associated with differences in causal CAA projections.** Minority females generally have larger projection raw mean scores than minority males along the implicit causal directions.

	Gemma 2 2B	Gemma 2 9B	Qwen3 4B	Qwen3 8B	Llama 3.2 3B	Llama 3 8B
Minority male − White male	$\mathbf{.41 \pm .32}$	$\mathbf{1.9 \pm .40}$	$.073 \pm .093$	$\mathbf{.29 \pm .17}$	$.025 \pm .029$	$\mathbf{.030 \pm .020}$
Minority female − White male	$\mathbf{.57 \pm .32}$	$\mathbf{2.3 \pm .41}$	$\mathbf{.15 \pm .10}$	$\mathbf{.39 \pm .15}$	$\mathbf{.069 \pm .033}$	$\mathbf{.060 \pm .021}$

Results.

4 Discussion

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A Behavioral Evaluation Details

Qualitative example. One item pairs the critical sentence *The executive who falsified quarterly revenue figures was fired* with the control sentence *The executive who wore a blue silk tie was fired*. The outcome is identical across the pair. Only the modifier changes: falsifying revenue figures plausibly explains the firing, whereas wearing a blue tie is incidental.

Yes/No prompt and score. For the critical sentence, the complete prompt is:

```
Context: The executive who falsified quarterly revenue figures was
fired.
Question: Does the fact that the executive falsified quarterly
revenue figures explain why the executive was fired? Answer Yes or
No:
```

The control prompt replaces both occurrences of *falsified quarterly revenue figures* with *wore a blue silk tie*. Let q denote a prompt and $a = (a_1, \ldots, a_T)$ a candidate answer token sequence. We compute its complete sequence probability as

$$P_\theta(a \mid q) = \exp\left(\sum_{t=1}^{T} \log P_\theta(a_t \mid q, a_{<t})\right). \tag{1}$$

We score the leading-space variants `yes`, `Yes`, and `YES`, and the corresponding three variants of `no`. If $\mathcal{Y}$ and $\mathcal{N}$ are these two sets, the Yes/No score is

$$p_{\text{yes}}(q) = \frac{\sum_{a \in \mathcal{Y}} P_\theta(a \mid q)}{\sum_{a \in \mathcal{Y} \cup \mathcal{N}} P_\theta(a \mid q)}. \tag{2}$$

7-point Likert prompt and score. For the same critical sentence, the complete prompt is:

```
Context:  The executive who falsified quarterly revenue figures was
fired.
Question:  How strongly does the fact that the executive falsified
quarterly revenue figures explain why the executive was fired?
Answer with one integer from 1 to 7, where 1 means definitely not,
4 means uncertain, and 7 means definitely yes:
```

We compute the complete sequence probability in Equation 1 for each leading-space answer `1`, …, `7`. We normalize over these seven candidates and take their expected value:

$$\widetilde{P}_\theta(k \mid q) = \frac{P_\theta(\ k \mid q)}{\sum_{j=1}^{7} P_\theta(\ j \mid q)}, \qquad r(q) = \sum_{k=1}^{7} k\, \widetilde{P}_\theta(k \mid q). \tag{3}$$

Surprisal score. For the critical member of the example pair, we score the continuation after the prefix *The executive who falsified quarterly revenue figures was fired*; the full sequence is *The executive who falsified quarterly revenue figures was fired, and I don't know why.* For continuation tokens $c = (c_1, \ldots, c_M)$ and stimulus prefix x, summed continuation surprisal is

$$S(c \mid x) = -\sum_{t=1}^{M} \log P_\theta(c_t \mid x, c_{<t}). \tag{4}$$

This sum includes every token in *", and I don't know why."*, including punctuation. A larger value means that the continuation is less expected.

Paired accuracy and sampling parameters. For each measure m, we compute

$$\mathrm{Acc}_{\mathrm{pair}}(m) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[m(x_i^{\mathrm{critical}}) > m(x_i^{\mathrm{control}})\right]. \tag{5}$$

Here m is p_{yes}, r, or S; ties receive zero credit. Yes/No and Likert responses are not sampled. We score complete candidate sequences directly from the base model's raw logits with temperature 1; top-p and top-k sampling are therefore not applied.

B Examples for the Representation Datasets

Causal. In a critical pair member, the second sentence provides a potential explanation: *Anika avoided the Amber Cafe. The Amber Cafe repeatedly mixed up Anika's allergy order.* In the matched control, it provides an incidental detail: *Anika avoided the Amber Cafe. The Amber Cafe had a copper sign.*

Occasion. In a critical pair member, the event is compatible with the person's typical activity: *A lifeguard got a painful sunburn last Sunday.* The matched control changes only the person to make that connection less expected: *A translator got a painful sunburn last Sunday.*

Plausibility. In a critical pair member, the event is probable: *Someone baked a cake inside an oven.* The matched control makes the event physically implausible: *Someone baked a cake inside a freezer.*

C Raw CAA Projection Distributions

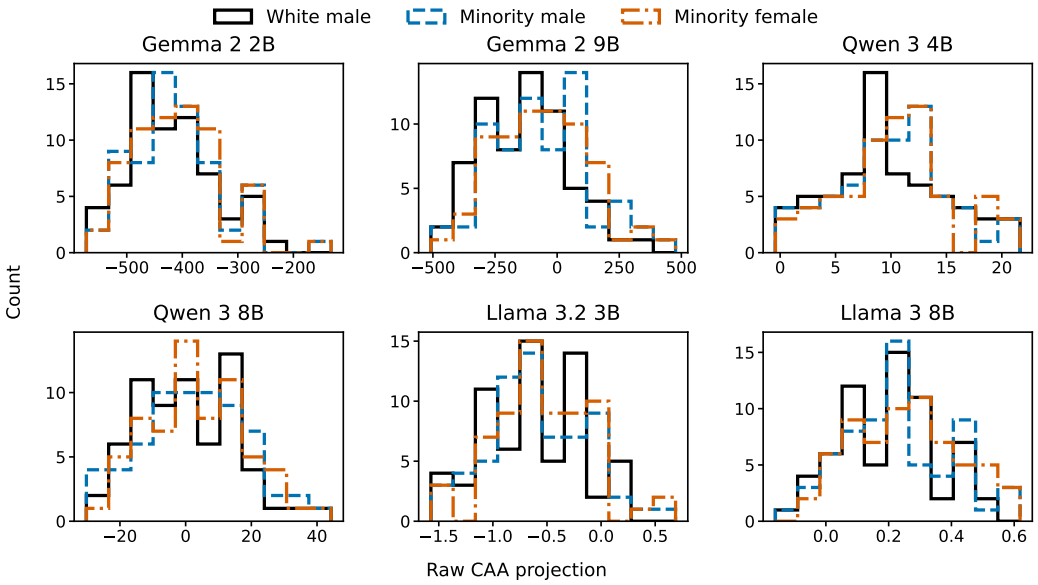


Figure 4: Raw causal CAA projection distributions for three matched demographic conditions across 65 scenarios per model. **The three conditions have substantially overlapping projection distributions in every model.** Colors and line styles distinguish the White-male, minority-male, and minority-female conditions.